IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MODELS FOR TECHNO-ECONOMICALLY OPTIMAL POWER SYSTEM EXPANSION

Friday, 2 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Answer all questions

1. **Variable renewable energy and storage**

a) The average wind generation profile at 3 potential sites, A, B & C for a 1GW wind farm are estimated by measurements of wind speed taken over a period of five years and are given in Table 1.

Table 1. Wind generation profiles at three potential sites A, B, and C.

	Midnight to 6AM	6AM to Noon	Noon to 6PM	6PM to Midnight
Wind site A (MWh)	1980	2040	2100	2280
Wind site B (MWh)	2040	2100	2280	1980
Wind site C (MWh)	2100	2030	2280	1860

i) Calculate the capacity factor of each of the three potential 1 GW wind farms. [2]

ii) These capacity factor calculations are based on many simplifying assumptions and are therefore likely under- or over-estimates of the true capacity factor. Indicate if they are under-estimates, over-estimates and briefly explain your answer with reference to the simplifying assumptions. [2]

iii) Calculate the yearly electrical energy that would be produced by each of these wind farms in TWh. [2]

iv) If the cost of constructing each wind farm is identical, rank the wind farms from highest to lowest LCOE. [2]

v) Without knowing the power system that these wind farms may be connected to identify which wind farm will be the least beneficial and explain your answer. [2]

vi) These LCOE calculations are also based on many simplifying assumptions and are therefore likely under- or over-estimates of the true LCOE. Indicate if they are under-estimates, over-estimates and briefly explain your answer with reference to the simplifying assumptions. [5]

b) The average demand profile for a regional power system is given in Table 2.

Table 2. Average demand in GWh

Midnight to 2AM	8.4
2AM to 4AM	8
4AM to 6AM	8.2
6AM to 8AM	11
8Am to 10AM	16
10AM to noon	18
Noon to 2PM	20
2PM to 4PM	26
4PM to 6PM	21
6PM to 8PM	17
8PM to 10PM	12.4
10PM to midnight	10

i) Calculate the yearly electrical energy demand for this regional power system in TWh.

[2]

ii) The three potential sites A, B & C from above are all in the same region as this power system. Indicate which 1 GW wind farm will be the most beneficial to build and connect to this regional power system and briefly explain your answer.

[6]

iii) Calculate the percentage of demand that will be supplied by the 1 GW wind farm at the most beneficial site.

[2]

c) In addition to the 1 GW of wind, the installed capacity includes 4 GW of nuclear and gas power plants. If the nuclear generation is inflexible and must run at 4 GW all year round, how much wind curtailment will be expected per year if the 1 GW wind farm at the most beneficial site is built and operated?

[8]

2. **Modelling industrial electricity demand**

a) In your basic capacity expansion model, demand for electricity is fixed and inflexible. As industry transforms their processes to reduce their greenhouse gas emissions, they are considering what capacity of equipment to buy and how much storage capacity they should build for their products. In simple terms, their equipment consumes electricity to produce goods.

i) Please list the decision variables for the industrial electricity demand. For each variable, provide a symbol, its units, and its description.

[5]

ii) Assume that the price of the product is fixed and that the warehouse for goods has costs to control the temperature in the warehouse and costs for loading and unloading goods. Please list five new terms that you would include in the objective function. For each term, provide the formulation and a short description.

[5]

iii) Assume that the demand for the product is fixed per day and that η products are produced per MW of equipment and hour of use. Please provide the formulation for five constraints that govern the operation of the electric equipment and the warehouse for the industrial demand. Provide a brief description (no more than a sentence) for each constraint.

[5]

iv) List all parameters you used in the objective function and constraints with their symbol, short description, and units.

[5]

b) Assume that your model is a two-stage stochastic program that includes multiple scenarios *s*. Each scenario has its own values for production of variable renewable energy resources and residential electricity demand. For each new decision variable you listed in 2.a.i, state if they will be first-stage or second-stage and write a sentence that justifies your answer.

[5]

c) Your colleague informs you that you only have demand data for the product on an annual basis. Based on their experience, your colleague anticipates that the demand for the product is either flat over the year or follows the variable renewable electricity generation. They know a company that sells hourly data.

i) Please draw a decision tree that can help you decide if you should buy the data.

[4]

ii) Would the value of information be higher, lower, or the same if you buy the data before or after you buy the electrical equipment and build the warehouse?

[2]

iii) Assume that the storage of goods is unlimited at zero cost. How different would the capital and operating expenses be between the two perfect foresight solutions (one perfect foresight solution for flat demand, one perfect foresight solution for variable demand)? Would the value of information be higher, lower, or the same if you buy the data before or after you buy the electrical equipment and build the warehouse? [3]

3. Investment plans under uncertainty

a) A system operator plans transmission investments considering three equiprobable scenarios. Table 3 shows the costs of perfect foresight transmission plans for each scenario. The operator also knows the probability-weighted cost of an integer solution that a stochastic mixed integer program found, which is £800, and the expected value of the expected value solution which is £900.

Table 3. Costs for perfect foresight plans per scenario (in £)

	PF- Plan 1	PF- Plan 2	PF-Plan 3
Scenario 1	700	10,000	10,000
Scenario 2	10,000	600	10,000
Scenario 3	10,000	10,000	500

i) Please calculate the expected value of the perfect foresight solutions. [2]

ii) Please provide an estimate for the MIP gap. Indicate if your estimate is a lower or an upper bound for the MIP gap. [2]

iii) Please estimate the Value of Stochastic Solution. Indicate if your estimate is a lower or an upper bound for the VSS. [4]

iv) Please estimate the Expected Value of Perfect Information. Indicate if your estimate is a lower or an upper bound for the EVPI. [4]

b) The table below has scenario-specific costs for three plans that were identified using models with different objective functions.

Table 4. Costs for three plans per scenario (in £)

	Plan 1	Plan 2	Plan 3
Scenario1	1300	1300	1200
Scenario 2	850	1200	1000
Scenario 3	850	1100	1000

i) Draw the cumulative risk profiles of the three plans (Plan 1, Plan 2, Plan 3). [3]

ii) Please explain if any out of the three plans dominate another of the three plans stochastically or deterministically.

[4]

iii) Indicate which solution a decision maker that minimizes the maximum regret would choose considering only information in Table 4.

[3]

iv) Which plan would the decision maker choose if they consider information from Tables 3 and 4?

[2]

v) Would you advise the decision makers to consider information from Table 3 when applying the min max regret criterion?

[3]

vi) Create a trade-off curve based on the information from the three plans in Table 4 that shows the trade-off between max cost across scenarios and expected cost across scenarios. [3]

vii) Which plan(s) would you advise the decision makers to present to the community and why? What are the benefits and challenges of presenting more than 1 plan to a community?

[3]

Handout from lecture notes

1. **Basic capacity expansion formulation including investment and operational decisions.**

   Objective function

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

   Decision variables:

   - nse_t: involuntary demand curtailment in MWh
   - cap_g: installed capacity in MW
   - $gen_{g,t}$: dispatch level of generator g at time t

   Constraints:

   Supply-demand balance: $\sum_g gen_{g,t} + nse_t = Demand_t \quad \forall t$

   Capacity constraint for each generator: $gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \; \forall g, t$

   Non negativity constraints: $nse_t,\ cap_g,\ gen_{g,t} \geq 0$

2. **Two power flow formulations**

   a) PTDF-based

   Objective function

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

   Decision variables:

   - nse_t: involuntary demand curtailment in MWh
   - cap_g: installed capacity in MW
   - $gen_{g,t}$: dispatch level of generator g at time t

   Constraints:

   Supply-demand balance: $\sum_g gen_{g,t} + \sum_n nse_{n,t} = \sum_n Demand_{n,t} \quad \forall t$

   Transmission line limits:

$$P_l^{MIN} \leq \sum_n PTDF_{l,n} \cdot \left(-Demand_{n,t} + \sum_g df_{g,n} \cdot gen_{g,t} + nse_{n,t}\right) \leq P_l^{MAX} \; \forall l, t$$

   Capacity constraint for each generator: $gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \; \forall g, t$

   Non negativity constraints: $nse_t,\ cap_g,\ gen_{g,t} \geq 0$

b) B-θ

Objective function

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

Decision variables:

- nse_t: involuntary demand curtailment in MWh
- cap_g: installed capacity in MW
- $gen_{g,t}$: dispatch level of generator g at time t
- $\theta_{n,t}$

Constraints:
Supply-demand balance: $\sum_g gen_{g,t} + \sum_n nse_{n,t} = \sum_n Demand_{n,t} \quad \forall t$

Supply-demand balance at nodal level:

$$\sum_g df_{g,n} \cdot gen_{g,t} + nse_{n,t} - \sum_l A_{l,n} \cdot \sum_{n'} B_l \cdot A_{l,n'} \cdot \theta_{n',t} = Demand_{n,t} \quad \forall n,\ t$$

Transmission line limits:
$P_l^{MIN} \leq \sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \leq P_l^{MAX} \ \forall l, t$

Capacity constraint for each generator: $gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \ \forall g, t$
Non negativity constraints: $nse_t,\ cap_g,\ gen_{g,t} \geq 0$

3. **Big-M reformulation of bilinear terms**
Constraint with bilinear term in transmission expansion problem:

$\sum_n B_l \cdot A_{l,n} \cdot \boldsymbol{\theta_{n,t}} \cdot \boldsymbol{u_l} = flow_{l,t} \quad \forall l_{candidate}$

Reformulation:

$-M \cdot (1 - u_l) \leq flow_{l,t} - \sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \leq M \cdot (1 - u_l) \quad \forall l_{candidate}$

M: big enough constant

4. KKT conditions

Stationarity: $\frac{\partial L}{\partial x} = \frac{\partial f_0}{\partial x} + \sum_{i=1:m} \lambda_i \cdot \frac{\partial f_i}{\partial x} + \sum_{i=1:p} v_i \cdot \frac{\partial h_i}{\partial x} = 0, \frac{\partial L}{\partial v} = 0$

Primal feasibility: $h_i(x) = 0, f_i(x) \leq 0$

Complementarity: $\lambda_i \cdot f_i(x) = 0$

Dual feasibility: $\lambda_i \geq 0$

5. **Lecture 3 problem formulation**

Objective function

$$\min \sum_{h \in H} \left(\sum_{g \in G} VarOM_g \cdot P_{g,h} + \sum_{r \in RES} VarOM_r \cdot R_{r,h} + VOLL \cdot NSE_h \right)$$

Constraints

Supply-demand balance: $\sum_{g \in G} P_{g,h} + \sum_{r \in RES} R_{r,h} + NSE_h = Demand_h \; \forall \; h$

Capacity constraint for thermal generator:

$0 \leq P_{g,h} \leq CAP_g \;\; \forall \; g, h$

Approximate inelastic operation of nuclear generation:

$P_{NUCLEAR,h} = CAP_{NUCLEAR} \;\; \forall \;\; h$

Capacity constraint for renewable generators:

$0 \leq R_{r,h} \leq CAP_r \cdot CF_{r,h} \qquad \forall \; r \in RES, h \in H$

Non-negativity constraint: $NSE_h \geq 0 \; \forall \;\; h$

6. **Lecture 4 problem formulation**

Objective function

$$\min \sum_{h \in H} \left(\sum_{g \in G} VarOM_g \cdot P_{g,h} + \sum_{r \in RES} VarOM_r \cdot R_{r,h} + VOLL \cdot NSE_h \right)$$

Constraints

Supply-demand balance:

$\sum_{g \in G} P_{g,h} + \sum_{r \in RES} R_{r,h} + \sum_{\boldsymbol{s \in S}} \left(\boldsymbol{DISCHARGE_{s,h} - CHARGE_{s,h}} \right) + NSE_h = Demand_h \; \forall \; h$

Capacity constraint for thermal generator: $0 \leq P_{g,h} \leq CAP_g \;\; \forall \; g, h$

Approximate inelastic operation of nuclear generation: $P_{NUCLEAR,h} = CAP_{NUCLEAR} \; \forall \;\; h$

Capacity constraint for renewable generators: $0 \leq R_{r,h} \leq CAP_r \cdot CF_{r,h} \qquad \forall \; r, h$

Limit of charge rate: $\boldsymbol{0 \leq CHARGE_{s,h} \leq CAP_s \qquad \forall \; s \in S, h \in H}$

Limit of discharge rate: $\boldsymbol{0 \leq DISCHARGE_{s,h} \leq CAP_s \;\; \forall \; s \in S, h \in H}$

Limit of non-synchronous generation (inspired by EirGrid)

$$\sum_{\boldsymbol{r \in RES}} \boldsymbol{R_{r,h}} + \left(\boldsymbol{DISCHARGE_{BESS,h} - CHARGE_{BESS,h}} \right) \leq \boldsymbol{75\% \cdot Demand_h \; \forall \; h}$$

Evolution of state of energy:

$$\boldsymbol{SOC_{s,h} = SOC_{s,h-1} + CHARGE_{s,h} \cdot Eff_s - \frac{DISCHARGE_{s,h}}{Eff_s} \quad \forall \; s \in S, h \in H \backslash 1}$$

State of energy limits: $\boldsymbol{0 \leq SOC_{s,h} \leq SOC_s^{max} \; \forall \; s \in S, h \in H}$

End-effects for state of energy: $\boldsymbol{SOC_{s,1} = SOC_{s,t_{end}} = 50\% \cdot SOC_s^{max} \; \forall \; s \in S}$

Non-negativity constraint: $NSE_h \geq 0 \; \forall \;\; h$